

Project Design Phase Proposed Solution Template

Date	01 October 2026
Team ID	SWTID-2026-2391
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	When records of several branches (ECE, EEE, CSE) are kept in a single ServiceNow table, ordinary role-based table access cannot restrict who may read, create, edit or delete individual records. Student details such as email and phone number can be seen or changed by users who should not have access, and there is no easy way to give each operation (READ, CREATE, WRITE, DELETE) its own permission.
2.	Idea / Solution description	Create a custom Institution Details table (u_institution_details) with a Branch field and protect it with ServiceNow Access Control Lists. The READ ACL is an advanced, script-controlled ACL: the script returns true for administrators and for users with role bb1 and false for everyone else. Separate ACLs control the other operations – CREATE requires role bb2, WRITE requires bb3 and DELETE requires bb4. The configuration is tested by impersonating the EEE User and the administrator.
3.	Novelty / Uniqueness	Combines user roles, table records, field values and a server-side ACL script in a single access-control design. Each CRUD operation has its own custom role (bb1–bb4), so permissions can be granted one step at a time. Administrator access is preserved. The whole solution is configuration inside ServiceNow – no external application, database or API is required.
4.	Social Impact / Customer Satisfaction	Protects the personal information of students and faculty, reduces accidental or unauthorized changes and deletions, and builds trust in how the institution handles data. Administrators gain clear, testable control over access, and authorized users see only the records they are meant to work with.
5.	Business Model (Revenue Model)	Proposed model: a one-time implementation fee for setting up the role/ACL framework for an organization, plus an optional annual support and access-audit subscription. The ACL configuration can be packaged as a reusable template for other institutions and departments.
6.	Scalability of the Solution	The same role + ACL + script pattern can be extended to more branches or departments, more custom tables and additional roles. Conditions on multiple field values can be added to the script, and the solution runs on the ServiceNow cloud platform, which scales with the number of users and records. Future work includes automated ACL test cases and reporting dashboards.